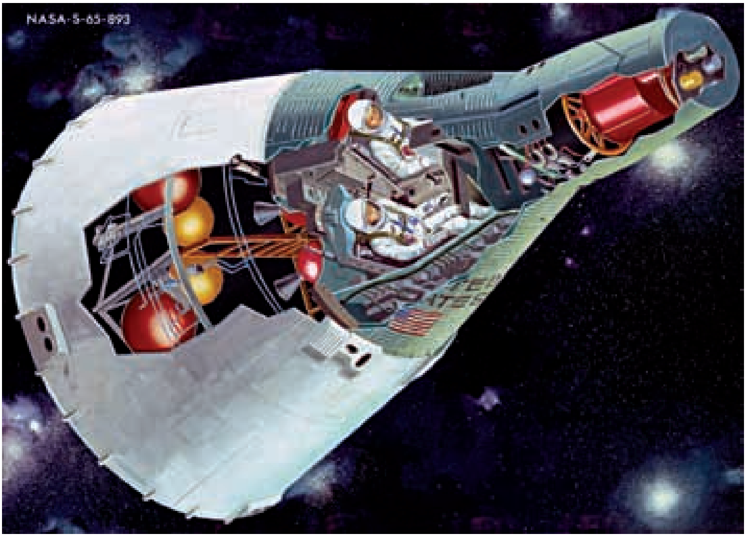
of holding two astronauts, seated side-by-side. The word gemini is Latin for "twins".

The Gemini spacecraft was an improvement on Mercury (it was originally called Mercury Mark II) in both, size and capability. Gemini weighed more than 3.5 tons – twice the weight of Mercury – but ironically seemed more cramped, having only 50 percent more cabin space for twice as many people. Ejection seats replaced Mercury's escape rocket, and more storage space was added for the longer Gemini flights. The long duration missions also required fuel cells instead of batteries for generating electrical power.

Unlike Mercury, which had only been able to change its orientation in space, Gemini needed real manoeuvring capability to rendezvous with another spacecraft. Gemini would have to move forward, backward and sideways in its orbital path, even change orbits. The complexity of rendezvous demanded two people on board, and more piloting than had been possible with Mercury. It also required the first on-board computers to calculate complicated rendezvous manoeuvres.

Gemini rode into orbit on a Titan II launch vehicle. The target for rendezvous operations was an the Agena – an unmanned spacecraft controlled by NASA, launched ahead of the Gemini. After meeting up in orbit, the



A cutaway illustration of the Gemini spacecraft